Assignment 1 - Git & GitHub Fundamentals

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1. Repository Setup
I used the given files and initialized a Git repository for the project folder using git init. I created a .gitignore file so that I can ignore local.env and other normal .log files while also keeping important.log from being ignored. I used git status so i can check whether what I'm doing is right, if my files are good, and how the state of the repository is.

2. README & Commit History
I created a README.md file that has my name, GitHub username, a project description, technologies used, a code example, a link, and the provided image. I did several commits so that I could record my progress, including changes to the README, HTML, CSS, and JavaScript files. I used git log –oneline –graph –all so I could see my commit history and inspect it.

3. Selective Staging
I changed index.html and style.css and then I used git add to stage index.html alone. I committed the file while leaving the changes to style.css unstaged. Then, I staged and committed style.css separately. This helped me understand that Git actually allows me the control to choose which changes should be included in a commit.

4. Branch & Merge
In this part, I created a branch called feature/about to add an About section to my README. I committed the change on that branch, switched back to main, and merged the feature branch into it.

5. Remote Changes

A commit saves changes in the local repository, while git push sends local commits to the remote repository on GitHub. Then git fetch is the one that updates Git's information about remote changes without integrating them into my current branch, while git pull downloads and integrates those changes.

To practice this, I edited my README directly on GitHub and committed the change there. I then used git fetch and checked my local branch was one commit behind the remote branch. Finally I used git pull to bring the change into my local repository.

6. Fair Use of AI

I used ChatGPT to help me with some troubles I had at the start with the files, as my VS code was not opening them as intended. Then, after fixing that problem, I also used it to help me understand Git commands and errors I've faced during the assignment. I asked for explanations when I was unsure what a command did and also how to fix a problem. I ran the commands myself and checked the results with git status constantly and then git log.

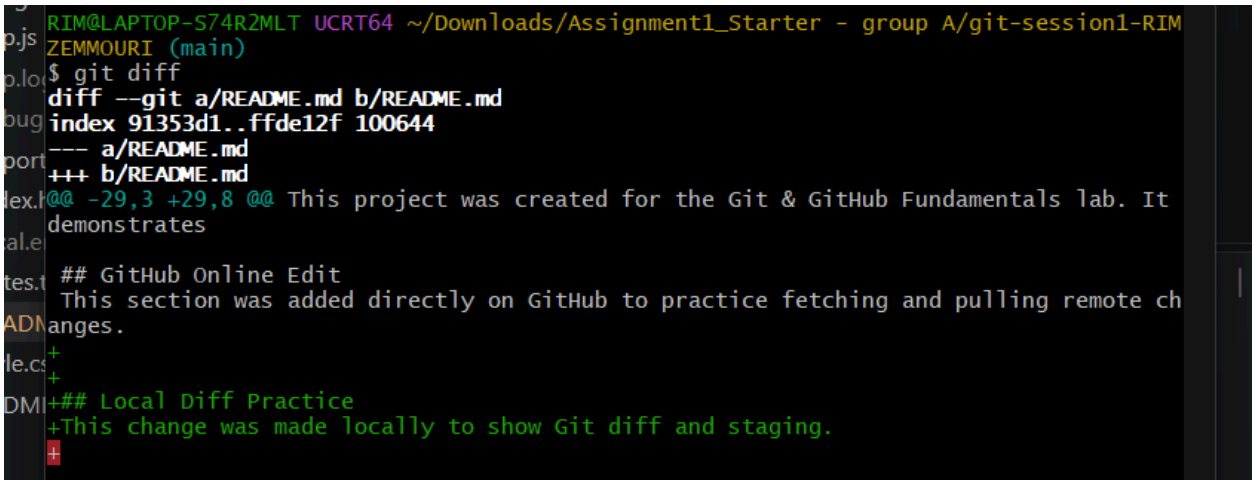
7. Sources & Reflection

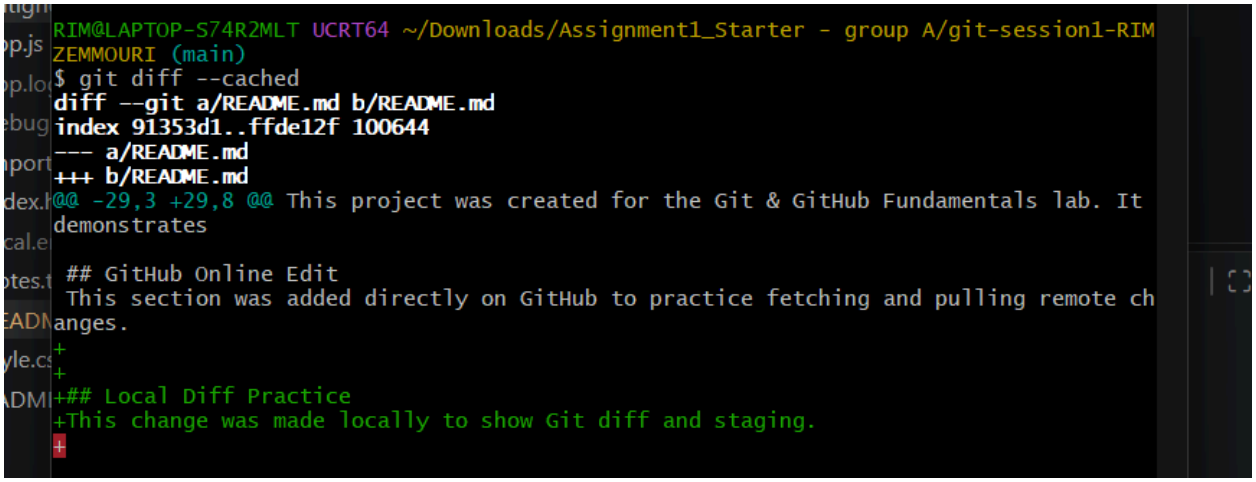
Sources

- Git documentation: https://git-scm.com/doc
- GitHub Docs: https://docs.github.com/en

The hardest concept for me was understanding the difference between staging, committing, and pushing changes. If Git behaves unexpectedly, I would first use git status to check my current branch and see which files are modified or untracked or staged. Using it helps a lot and made me pinpoint exactly what I should do next.

SCREENSHOTS GIT DIFF AND GIT DIFF –CACHED:




RIM@LAPTOP-S74R2MLT UCRT64 ~/Downloads/Assignment1_Starter - group A/git-session1-RIM
ZEMMOURI (main)
$ git diff --cached
diff --git a/README.md b/README.md
index 91353d1..ffde12f 100644
--- a/README.md
+++ b/README.md
@@ -29,3 +29,8 @@ This project was created for the Git & GitHub Fundamentals lab. It
demonstrates
GitHub Online Edit
This section was added directly on GitHub to practice fetching and pulling remote ch
anges.
+
+
+## Local Diff Practice
+This change was made locally to show Git diff and staging.
+